

Deliverable 3: Requirements Specification

Students are required to identify, analyze, classify, and document the requirements of their proposed system under the following three categories:

Business Requirements	Functional Requirements	Non-Functional Requirements
What does the business/stakeholder want to achieve?	What should the system do?	How well should the system perform?
Focus on business goals and objectives	Focus on system functions/features	Focus on quality attributes and constraints
Why is the system needed?	What will the system do?	What standards should the system meet?

1. Business Requirements

These describe the **business goals, problems, needs, and expected outcomes**.

Examples:

- The hospital wants to reduce manual appointment booking.
- The organization wants to reduce customer waiting time.
- Management wants to maintain digital records.
- The business wants to improve customer satisfaction.

2. Functional Requirements

These describe the **specific functions that the system must perform**.

Examples:

- The system shall allow users to register an account.
- The system shall allow patients to book appointments.
- The system shall allow administrators to manage doctor schedules.
- The system shall generate appointment reports.

3. Non-Functional Requirements

These describe **quality attributes, performance expectations, and constraints**.

Examples:

- The system should respond within 3 seconds under normal load.
- Only authorized users shall access patient records.
- The system should be available 99% of the time.
- The system should support 500 concurrent users.
- The interface should be responsive on desktop and mobile devices.

How should students perform this task?

Students should follow this sequence:

Requirement Elicitation → Requirement Analysis → Classification → Documentation → Validation

Step 1: Collect Requirements

Students should first collect information from relevant stakeholders using techniques such as:

- Interviews
- Questionnaires/Surveys
- Observation
- Meetings/Discussions
- Focus Groups
- Existing System Analysis
- Document Analysis

The **primary source should preferably be the actual stakeholder/user**.

Step 2: Identify Business Requirements

Ask:

What problem does the organization/business want to solve?

and

What does the organization want to achieve?

Convert the answers into clear business requirements.

Step 3: Derive Functional Requirements

For each business requirement, ask:

What functionality is required in the system to achieve this business objective?

For example:

Business Requirement:

Hospital wants to reduce manual appointment booking.

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Functional Requirement:

The system shall allow patients to book appointments online.

Step 4: Identify Non-Functional Requirements

For the proposed functionality, ask:

What quality, performance, security, usability, reliability, or scalability requirements should the system satisfy?

For example:

Functional Requirement:

Patient can book an appointment.

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Non-Functional Requirement:

The appointment booking operation should be completed within 3 seconds under normal system load.

Sources / Support for Requirements

Students can obtain support from:

1. Stakeholders / Real Users

This should be the **preferred source**.

Students can conduct:

- Interviews

- Surveys
- Questionnaires
- Observation
- Meetings

They should provide evidence such as interview questions, questionnaire results, meeting notes, or screenshots where appropriate.

2. Domain Experts

For specialized projects, students can consult a relevant domain expert.

For example:

- Healthcare → Doctor/Hospital representative
- Banking → Banking professional
- Education → Teacher/Academic administrator
- E-commerce → Business owner
- Agriculture → Agriculture/domain expert

3. Existing Systems and Documentation

Students may study:

- Existing systems
- Competitor applications
- User manuals
- Organizational documents
- Government/regulatory documents
- Industry standards
- Relevant research papers

However, **copy-pasting requirements from an existing system should not be accepted.** Students must explain why a requirement is relevant to their own project.

4. AI Tools

AI can be used as a **supporting tool**, but it should not be considered the primary source of requirements.

A good workflow is:

**Stakeholder → Requirement → AI-assisted analysis/classification → Student verification
→ Supervisor/Domain Expert validation**

Deliverable 3 Rubric — 10 Marks

Assessment Criteria	Marks
Business Requirements — clarity, relevance, and alignment with business objectives	4.0
Functional Requirements — correctness, completeness, and clarity	7
Non-Functional Requirements — relevance, measurability, and quality attributes	4.0
Requirement Elicitation Evidence / Sources	2.0
Requirement IDs, organization, and documentation quality	1
Validation, consistency, and feasibility	2.0
Total	20.0